

Hello everyone. I am Sania Sadaf and welcome back to my AWS machine learning Engineering series. This presentation begins Domain 2, which covers machine learning models and foundation model development.

I will explain how an ML engineer chooses an appropriate model or AI service for a business problem. The choice may involve a traditional machine learning algorithm, a foundation model from Amazon Bedrock, a managed AWS AI service, or a custom solution. I will also compare prompting, Retrieval Augmented Generation, and fine-tuning. The goal is to understand the decisions behind model selection without using code or performing a live experiment.

The main idea is that the most advanced model is not automatically the best option. A useful solution must fit the business task, available data, required quality, response time, budget, privacy rules, and the team's ability to operate it.

A model choice should begin with a clear definition of the problem. The first question is what the system needs to produce. Some systems predict a number, classify an item, detect unusual behavior, recommend a product, generate text, summarize documents, answer questions, or extract information.

The next consideration is the data. The engineer should know whether labeled examples exist, how much data is available, which languages and formats appear, and whether the data represents real use.

Operating limits also affect the choice. A medical or financial decision may require stronger explanations and human review. A real-time application may require low latency, while a background process can accept a slower model.

Traditional machine learning includes several learning setups. Supervised learning uses labeled examples.

Unsupervised learning works without a labeled target. Clustering can group similar customers or documents, while anomaly detection can identify behavior that differs from normal patterns.

Time-series models use the order of observations to predict future values.

Recommendation systems use past behavior, item details, and user context to suggest relevant choices.

Foundation models are large pretrained models that can support many tasks. Amazon Bedrock provides access to foundation models from different providers. Choosing among them requires more than comparing general model rankings.

Capability fit comes first. The model should support the required task, language, input type, context length, and output format.

Quality fit means checking accuracy, relevance, consistency, reasoning behavior, and safety for the real use case. Operational fit includes latency, throughput, regional availability, customization options, and price.

Organizations can build a custom model, use a managed service, or adapt a pretrained model. A custom solution offers greater control over algorithms, features, and training. It may fit a specialized problem, but it usually requires more data, machine learning knowledge, infrastructure, testing, and long-term maintenance.

Managed AWS AI services can solve common tasks quickly. Amazon Textract extracts information from documents. Amazon Rekognition supports image and video analysis. Amazon Comprehend analyzes text, and Amazon Transcribe converts speech into text.

Pretrained models and foundation models provide a middle path. The team can begin with an existing model and use prompts, retrieval, or fine-tuning to adapt it. The best decision depends on how unique the task is and how much control the organization needs

Prompting, Retrieval Augmented Generation, and fine-tuning solve different types of gaps. Prompt engineering improves the instructions given to a foundation model.

RAG connects a model to external knowledge. The system retrieves relevant documents and gives that content to the model as context.

Fine-tuning changes model behavior by training selected examples. It can help with repeated formats, specialized language, style, or task behavior.

These methods can also work together. A system may use a strong prompt, retrieve current information, and use a fine-tuned model.

AWS AI services provide ready-made capabilities for common business problems. Amazon Textract can identify printed text, handwriting, forms, and tables in documents. ]

Amazon Comprehend performs natural language processing tasks such as sentiment analysis, entity detection, and key phrase extraction. Amazon Rekognition analyzes images and video. Amazon Transcribe converts speech into text, which can then support search, summaries, or further language analysis.

To conclude Part 1, model selection should follow a repeatable decision process. First, define the required output and identify which mistakes are most important. Then review the available data, labels, languages, and modalities. These details narrow the possible solutions.

That is all for this part, thank you for watching and see you in the next video.